Counting Necklaces

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2209 | Mathematics | https://cses.fi/problemset/task/2209

Your task is to count the number of different necklaces that consist of n pearls and each pearl has m possible colors.

Two necklaces are considered to be different if it is not possible to rotate one of them so that they look the same.

Input

The only input line has two numbers n and m: the number of pearls and colors.

Output

Print one integer: the number of different necklaces modulo 10^9+7.

Constraints

- $1 \le n,m \le 10^6$

Example

Input
4 3

Output
24